Example 6.13

The table below gives the prices of base year and current year of 5 commodities with their quantities. Use it to verify whether Fisher's ideal index satisfies time reversal test.

Commodity	Base year		Current year	
	Unit price (₹)	Quantity	Unit price (₹)	Quantity
A	4	40	5	60
B	5	50	10	70
C	8	65	12	80
D	6	20	6	90
E	7	30	10	75

Solution:

Index number by Fisher's ideal index method

Commodity	p_0	q_0	p_1	q_1	p_0q_0	p_0q_1	p_1q_0	p_1q_1
A	4	40	5	60	160	240	200	300
B	5	50	10	70	250	350	500	700
C	8	65	12	80	520	640	780	960
D	6	20	6	90	120	540	120	540
E	7	30	10	75	210	525	300	750
					1260	2295	1900	3250

$$P_{01} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}}$$

$$= \sqrt{\frac{1900}{1260} \times \frac{3250}{2295}}$$

$$P_{10} = \sqrt{\frac{\sum q_0 p_1}{\sum q_1 p_1} \times \frac{\sum q_0 p_0}{\sum q_1 p_0}}$$

$$= \sqrt{\frac{2295}{3250} \times \frac{1260}{1900}}$$

Hence, $P_{01} \times P_{10} = \sqrt{\frac{1900}{1260} \times \frac{3250}{2295} \times \frac{2295}{3250} \times \frac{1260}{1900}}$

$$= \sqrt{1} = 1$$

Fisher's Index number satisties time reveral test.

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